Learning SPARQL by Example

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Abstract

This half-day tutorial provides a hands-on introduction to SPARQL, the query language for RDF-based semantic data. It is designed for both beginners and intermediate users, requiring no prior SPARQL knowledge. Through a learning-by-example approach, participants are introduced to two complementary learning and documentation infrastructures: *sparql-examples*, a standardised SPARQL metadata format developed by the SIB Swiss Institute of Bioinformatics, and *SPARQL-Fest*, a new front-end platform that collects, publishes, and showcases SPARQL queries produced during workshops.

After an overview of SPARQL concepts and tools such as Wikidata, participants work in small groups to formulate questions answerable via SPARQL, explore RDF resources, design and document their queries, and publish them using the SPARQL Example format through the SPARQL-Fest GitHub workflow. The tutorial also presents several SIB tools supporting SPARQL learning, including the SPARQL Playground and SPARQL Guidelines. By the end of the session, attendees will have created a fully documented SPARQL query, contributed to a shared knowledge base, and provided feedback on the initial deployment of the SPARQL-Fest platform.

Keywords

SPARQL, Semantic Web, FAIR data, Life Sciences, Health Care, Query documentation

1. Introduction

SPARQL is the W3C-standard query language for the Resource Description Framework (RDF) and plays a central role in semantic data integration across healthcare and life sciences. Despite its importance, SPARQL often presents a steep learning curve due to the need to understand both the query language and the underlying data models, schemas, and metadata.

This tutorial proposes a structured, example-driven introduction to SPARQL, delivered as a half-day session at SWAT4HCLS 2026. The tutorial combines short lectures, demonstrations, and supervised group work, allowing participants to actively engage with real-world RDF data and contribute reusable SPARQL examples to a public repository.

2. Audience and Required Skills

2.1. Intended Audience

The tutorial targets researchers, data stewards, PhD students, ontologists, and developers working with semantic technologies in life sciences and health care.

2.2. Required Skills

Participants are expected to bring a laptop and have access to a GitHub account. Basic familiarity with RDF or SPARQL is helpful but not required. The tutorial is designed to remain accessible to beginners

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discovering SPARQL for the first time, while also offering value to more experienced users wishing to refine their skills or share knowledge within a community.

3. Tutorial Overview and Learning Objectives

The tutorial follows a learning-by-example approach and is built around two complementary infrastructures:

- **sparql-examples**: a GitHub-based repository defining a standardised SPARQL metadata format, developed by the SIB Swiss Institute of Bioinformatics, enabling consistent documentation, validation, and reuse of SPARQL queries, including SHACL-based quality control [**?**].
- **SPARQL-Fest**: a front-end platform currently under development that aggregates SPARQL workshop outputs, showcases participant-contributed queries, and supports a structured learning experience based on real-world examples.

By the end of the tutorial, participants will:

- understand the importance of SPARQL in healthcare and life sciences,
- create and publish a fully documented SPARQL query,
- contribute to a growing community knowledge base,
- learn to navigate SIB tools for semantic data exploration,
- interact with peers sharing similar interests,
- provide feedback on SPARQL-Fest and related SPARQL learning tools.

4. Timeline and Structure

The tutorial is designed as a half-day session of approximately 3.5–4 hours:

- Introduction and session overview (15 min)
- What is SPARQL and why it matters in health and life sciences (20–30 min)
- SIB tools for learning SPARQL (15–20 min)
- Demonstration of the complete workflow by example (15–20 min)
- Reflection time (5–10 min)
- Coffee break
- Group work: designing, documenting, and publishing queries (60–90 min)
- Final discussion and feedback (20–30 min)

5. Detailed Session Description

5.1. Introduction and Session Overview

The session begins with a presentation of the tutorial plan and an introduction to the SPARQL-Fest platform, designed to host workshop events, collect participants' SPARQL queries, and provide a shared learning space. The SPARQL Example standard is introduced, including its motivations, structure, and benefits for reusable teaching material. The facilitators and their affiliations are also presented.

5.2. What Is SPARQL and Why It Matters

This section introduces SPARQL, the problems it addresses, and its role as a key technology for semantic data integration. Fundamental SPARQL concepts are explained using clear examples drawn from healthcare, biomedical datasets, and FAIR data ecosystems. A complete SPARQL query is demonstrated and interpreted step by step.

5.3. SIB Tools for Learning SPARQL

Participants are introduced to several tools developed or supported by the SIB Swiss Institute of Bioinformatics, including:

- SPARQL Guidelines and best practices for example usage,
- the SPARQL Example format and its metadata fields,
- SHACL-based validation using `sparql-examples-utils.jar`,
- the SPARQL Playground for interactive query experimentation.

5.4. Demonstration of the Complete Workflow

A complete end-to-end workflow is demonstrated: identifying a question answerable via SPARQL, locating relevant RDF resources, exploring namespaces, classes, and properties, drafting and refining a query, documenting it using the SPARQL Example format, publishing it on GitHub, and visualising it through the SPARQL-Fest interface.

5.5. Group Work

After a short reflection period and coffee break, participants form small groups based on shared interests. Each group follows the demonstrated workflow to design, document, and publish a SPARQL query. Facilitators circulate to provide guidance, and short checkpoint moments are used to share interesting challenges, solutions, or insights with the entire group.

5.6. Final Discussion and Feedback

Each group briefly presents their query, the question they explored, and the challenges encountered. Participants provide feedback on SPARQL-Fest and suggest ideas for future SPARQL learning tools or improvements. This session also serves as an initial evaluation of the SPARQL-Fest platform by real users.

6. Previous Editions

This is the first edition of the tutorial in this integrated format. Individual components have previously been used independently for educational purposes, but not as a unified workflow combining SPARQL Examples and SPARQL-Fest.